Fall-Risk Monitoring in Diverse Terrains Using Dual-Task Learning and Wearable Sensing System

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***Abstract*—As the elderly population grows, falling accidents become more frequent, and the need for fall-risk monitoring systems increases. Deep learning models for fall-risk movement detection neglect the connections between the terrain and fall-hazard movements. This issue can result in false alarms, particularly when a person encounters changing terrain. This work introduces a novel multisensor system that integrates terrain perception sensors with an inertial measurement unit (IMU) to monitor fall-risk on diverse terrains. Additionally, a dual-task learning (DTL) architecture that is based on a modified CNNLSTM model is implemented; it is used to determine fall-risk level and the terrain from sensor signals. Three fall-risk levels – "normal," "near-fall," and "fall" - are identified as being associated with "flat ground," "stepping up," and "stepping down" terrains. Ten young subjects performed 16 activities on flat and stepping terrains in a laboratory setting, and ten elderly individuals were recruited to perform four activities in the hospital. The accuracies of classification of fall-risk levels and terrains by the proposed system are 97.6% and 95.2%, respectively. The system detects pre-impact fall movements, with a fall prediction accuracy of 97.7% and an average lead time of 329ms for fall trials, revealing the model's effectiveness. The overall monitoring accuracy for elderly individuals is 99.8%, confirming the robustness of the proposed system. This work discusses the impact of sensor type and the model architecture of DTL on the classification of fall-risk levels across various terrains. The results demonstrate that the proposed method is reliable for monitoring the risk of falling.**

***Index Terms*—Fall-risk monitoring, pre-impact fall detection, terrain classification, wearable devices, deep learning, dual-task learning.**

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the National Cheng Kung University Hospital Institutional Review Board (Human Experiment Review Committee) under Application No. A-ER-110-211, and performed in line with the Declaration of Helsinki.

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I. Introduction

As THE population ages, the incidence of falling accidents among the elderly is growing [1]. The main causes of falling accidents for the elderly are the natural deterioration of physical function or the encountering of a fall-hazard environment during daily activities [2], [3], [4]. The impact of a fall generates a large force, imposing a significant risk of injury or even fatality. Fall detection systems facilitate the timely assistance of individuals who have fallen [5], [6], [7], [8], [9], [10]. However, an injury that is caused by the impact of a fall can have long-term negative effects, such as chronic pain, the need for costly high healthcare, and a heightened likelihood of subsequent falls [1], [2]. To safeguard the elderly from fall-related injuries, environmental hazards must be identified, and fall-risk movements that are associated with impairments and disabilities of the elderly must be assessed.

Mobility and gait are key indicators of the risk of falling for the elderly [11], [12], [13], [14], [15], [16], [17], [18]. Some works have demonstrated that gait analysis during elevation changes can help to evaluate how the elderly respond to changes in terrain [16], [17], [18]. By detecting abnormal gait or unstable posture on different terrains, the risk of falling can be inferred. However, such gait assessments do not suffice to evaluate balance capabilities during falling events and may not prevent falls. Therefore, pre-impact fall detection (PIFD) systems and wearable airbags have been developed to protect the elderly from the impact of falling [19], [20], [21]. To detect pre-impact falling movements, wearable devices that are integrated with inertial measurement units (IMUs) have been used [22], [23], [24], [25], [26], [27], [28], [29], [30], [36], [37], [38], [39]. IMUs measure the acceleration and angular velocity of a user's body, and the signals thus obtained can be used to identify the period of free-fall before impact on the ground. The free-fall period, called the pre-impact falling period, is critical in determining whether a fall has occurred. This period directly reflects the likelihood of body imbalance and falling, allowing a PIFD system to detect a fall before impact accurately.

Deep learning (DL) algorithms have emerged as advanced means of PIFD. A convolutional neural network (CNN) consists of convolutional layers that apply filters to sequential signals for feature extraction. By utilizing IMU signals as inputs to the CNN model, pre-impact fall movements can be distinguished

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from activities of everyday living (ADLs) [27]. A multi-source CNN ensemble (MCNNE) structure, which incorporates both IMU sensors and pressure sensors in a smart insole system, has been used to detect the direction of falling [28]. Sensor signals of each type are fed into a separate CNN model to extract sensor-based features. The ensemble model within the MCNNE uses these sensor-based features to classify falls and ADLs. To assess the robustness of such a PIFD system, experiments that involve near-fall movements must be conducted [29], [30], [31], [32]. Near-fall movements are highly similar to actual fall movements and should be correctly classified by the DL model to avoid false alarms. Accordingly, a modified directed acyclic graph-convolutional neural network (DAG-CNN) architecture has been proposed [30]. The DAG-CNN architecture simultaneously extracts multi-scale features - ranging from coarse to fine - from multiple layers. This architecture better classifies falls, near-falls, and ADLs, than traditional CNN architectures.

Unlike CNN-based methods, long short-term memory (LSTM) is a DL network that is designed for sequential data processing and is effective for tasks that require capturing long-term dependencies in signals [33], [34], [35], [36], [37], [38], [39]. To leverage the strengths of both CNN and LSTM, a hybrid method that combines CNN and LSTM networks has been implemented to improve the accuracy of PIFD [36], [37], [38], [39]. However, previously proposed DL models for PIFD are single-task learning (STL) models, which only classify movements by risk of falling. These DL models ignore the relationships between terrain and movement, so changes in terrain lead to false alarms. Moreover, falls resulting from environmental factors, such as tripping over obstacles or stairs, pose a higher risk of injury than others. Therefore, simultaneously classifying a user's movements and the terrain is crucial.

This work offers a terrain-robust method for monitoring the risk of falling with PIFD on various terrains, which presents three novel contributions to fall prediction and detection research. (a) This work is the first to develop a multisensor system that combines terrain perception sensors with IMUs for PIFD across diverse terrains. (b) It introduces a dual-task learning (DTL) architecture to simultaneously detect the level of fall risk associated with a movement and the terrain. This approach improves the model's ability to establish connections between the surrounding terrain and a user's movements during training. Therefore, this work achieves higher fall-risk level and terrain classification accuracy than previous methods. (c) The proposed model classifies movements into "normal," "near-fall," and "fall." These classifications refer to body balance on different terrains, providing caregivers with a tool for monitoring and reducing the risks of falling to which elderly individuals are subject on various terrains.

II. Materials and Methods

A. System Design

Fig. 1 presents the proposed multisensor system with sensor placements and the printed circuit board of two wearable

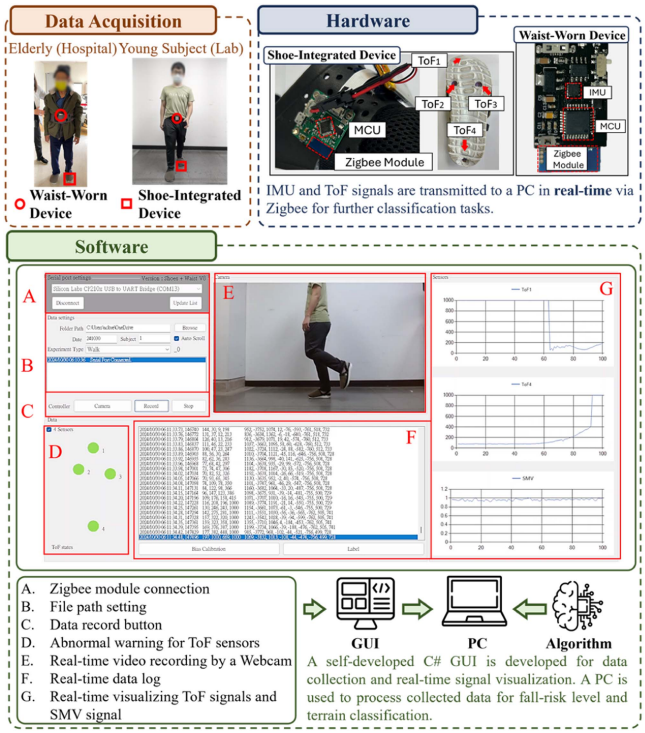


Fig. 1. Architecture of proposed multisensor system with data collection GUI for AI model development.

devices. The shoe-integrated device, modified from one in a previous work [10], consists of a microcontroller (Arduino ATmega328p, Atmel LLC), four Time-of-Flight (ToF) sensors (VL53l0x, STMicroelectronics), and a ZigBee module (DL-LN33, Soarland & Hexin Technology). The ToF sensors are used to measure distances between the sensors and targets in the sensing direction, facilitating the prediction of upcoming terrain [10], [40], [41]. Systems that incorporate ToF sensors into footwear or prosthetic legs have been shown to be able simultaneously to capture both the geometry of the terrain and the gait features of a walking user. In this system, the ToF sensor on the front of the shoe's outsole identifies obstacles or uneven terrain. The other three ToF sensors are embedded in the outsole under the heel and the ball region of the foot to determine the distance between the ground and the outsole of the shoe.

The IMU is attached at the waist of the body, which aligns with existing works [24], [25], [27], [30]. The waist-worn device is integrated with an IMU (MPU-9250, TDK InvenSense) and based on the same microcontroller and Zigbee module as the shoe-integrated device. IMUs capture acceleration and angular velocity through the integrated accelerometers and gyroscopes within the device. The scales of the acceleration and angular velocity in the IMU are ± 8g and $\pm 2000^{\circ}$/second, respectively. The signals of both the IMU and the ToF sensors are transmitted to a computer as a packet with a sampling rate of 30 Hz via the Zigbee module. The collected signals are processed by a computer that is equipped with an NVIDIA RTX 4080 GPU with 16 GB of memory and an Intel Core i7-13700 CPU for classifying fall-risk level and terrain.

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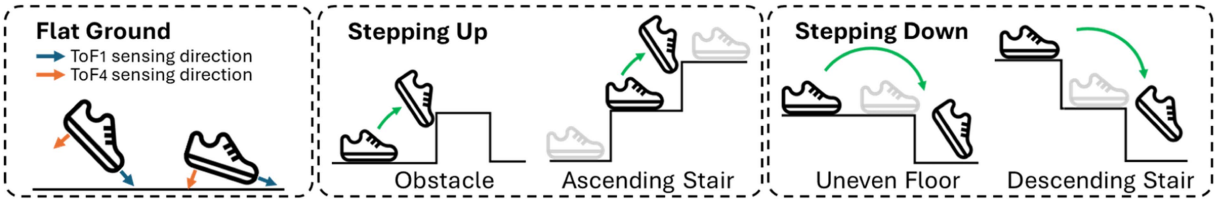


Fig. 2. Three types of terrain: flat ground representing a typical indoor environment; stepping up terrain, including obstacles and ascending stairs, and stepping down terrain, including uneven floors and descending stairs.

B. Experimental Protocol

This work collected 16 activities, which were conducted in the lab by ten young subjects, and four activities, which were performed by ten elderly subjects in a hospital setting. The young subjects were 24.4 ± 1.2 years old, with heights of 175 ± 5.2 cm and weights of 68.6 ± 8.2 kg. All the young subjects were male and had no health problems or disabilities. Three female and seven male elderly were recruited, aged 71.2 ± 7.3 years old, with heights of 160.9 ± 6.7 cm and weights of 65.2 ± 10.8 kg. All elderly individuals scored above 90 on the Barthel Index [42], indicating they can perform activities of daily living without assistance. Fig. 2 illustrates the types of terrains included in this work, which were selected based on previous research [43], [44], [45], [46]. Young subjects simulated five ADLs, four near-fall movements, and seven falls. The ADLs are walking, ascending stairs, descending stairs, sitting, and leg-shaking. Leg-shaking requires seated subjects to shake the shoe-integrated device to generate variable ToF signals while the IMU signal remains stable. In contrast, when subjects sit down, the vertical acceleration that is caused by sitting is captured by the IMU, but the ToF signals remain unchanged because the position of the feet is fixed. These activities are carried out to demonstrate that the model is not biased towards a single sensor type in classifying terrain and fall-risk levels. Seven types of falls are (i) forward faint falls, (ii) leftward faint falls, (iii) rightward faint falls, (iv) trip falls while crossing obstacles, (v) trip falls while ascending stairs, (vi) misstep falls on an uneven floor, and (vii) misstep falls while descending stairs. The height of the stairs and the obstacle is 15 cm. The fall trials are conducted using 15 cm-thick mattresses to protect subjects from injury. The near-fall trials are designed to simulate the early stages of falls that are caused by environmental factors, but subjects subsequently perform compensatory balancing reactions. Four near-fall trials are conducted on terrain with changing elevations; they are (i) tripping over an obstacle; (ii) tripping while ascending stairs; (iii) missteps on an uneven floor; and (iv) missteps while descending stairs. Each activity is executed six times, yielding a simulated dataset of 300 ADLs, 240 near-fall movements, and 420 falls.

The experiment that includes the elderly individuals involves walking on level surfaces, ascending and descending stairs, and sitting - all in a typical hospital environment. Due to the risk of injury, the elderly subjects do not simulate falling accidents or near-fall movements, and a total of 75 sequences are collected. The second group of experiments aims to monitor the elderly subjects' mobility and reactions on different terrains. The experimental protocols were approved by the Institutional Review Board of the NCKUH (A-ER-110-221).

C. Data Collection

Each data packet transferred from the wearable devices to the PC includes a timestamp generated by the microcontroller timer within the devices. This packet design enables the alignment of sensor signal collection times across wearable devices and helps identify any lost data during transmission. To conduct integrity checks, a real-time data visualization GUI has been developed using C# for data collection, as shown in Fig. 1. The GUI enables real-time monitoring of sensor values alongside video feedback, allowing researchers to identify any corrupted data promptly. In addition, ToF1, ToF4, and SVM signals are visualized in real-time in the GUI. Those signals are used to label the data for terrain and fall-risk levels; therefore, timely showing of the signals with video supports the researcher in checking the data quality. The GUI also logs the computer's datetime format and real-time data to validate signal integrity. This work manually reviewed all collected data to identify and remove sequences containing lost samples or corrupted data, ensuring the overall reliability of the dataset. Young subjects were asked to re-conduct experiments for the removed sequences to ensure there was enough training data for each class type. For elderly individuals, the number of re-conducted trials was adjusted to accommodate individual willingness. Consequently, the missing data from elderly individuals was removed, and the rest of the data was used for model training and testing.

D. Sensor Calibration and Signal Preprocessing

The sensors are calibrated to ensure sensor data quality. Ranging offset calibration is performed on ToF sensors to measure precise distance by compensating for inherent initial offset errors. The signals from the accelerometer go through a six-position static calibration [47], and the angular velocity from the gyroscope is calibrated using a zero-rate offset method [48]. The sensors' calibrations mitigate real-time sensor errors and error accumulation in the system and enable correct sensor data to be used for model training.

To calculate the magnitude of the acceleration on the IMU, a single magnitude vector (SMV) of tri-axial accelerations is calculated using the following equation:

$$SMV = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

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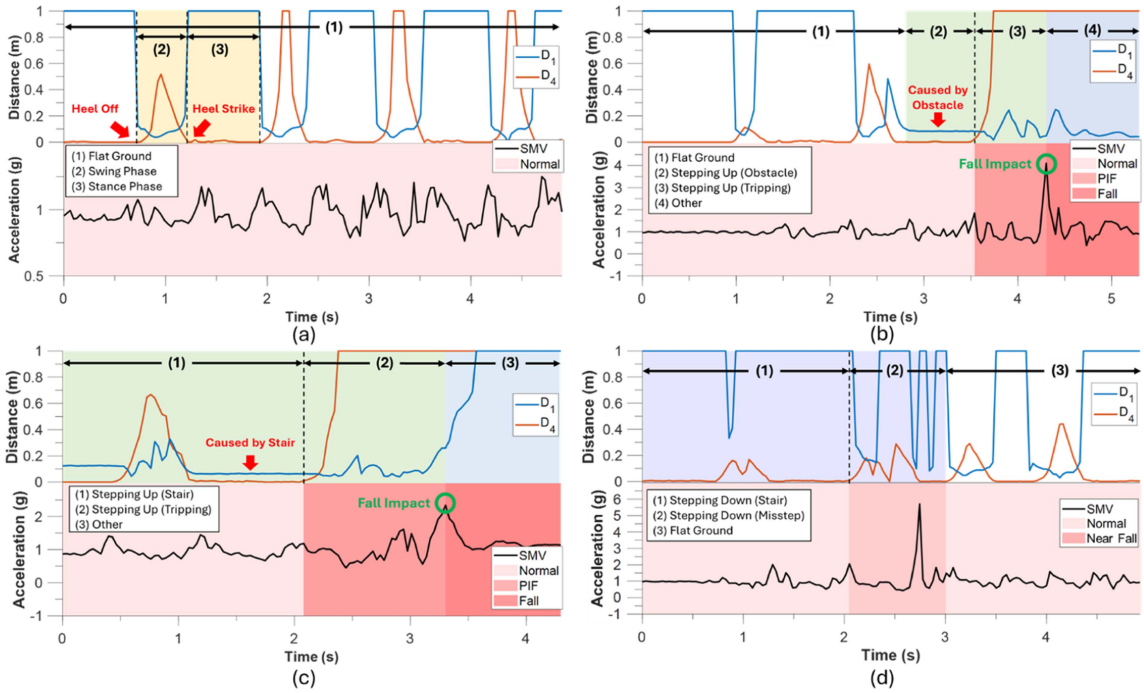


Fig. 3. Visualization of four activities using ToF signals, SMV signal, terrain label, and fall-risk level label. (a) Walking on flat ground; (b) walking and tripping fall; (c) ascending stairs and tripping fall; (d) descending stairs, misstep, and return to flat ground. Time of fall impact is indicated by a green circle in SMV signal. Pre-impact fall is denoted as PIF.

where a_x, a_y, and a_z represent the accelerations along the three axes. The SMV signal is one of the inputs for the proposed model. The ToF signals are denoted as D_i, where $i \in \{1, 2, 3, 4\}$ identifies the four ToF sensors. To detect the terrain geometry one step ahead of the shoe, the maximum range of the ToF signal is set to 1 m. Each sequence's ToF, IMU, and SMV signals are segmented using a sliding window. The segmented signals are used as inputs to train the model. The size of the window is determined by the duration of the gait cycle, which is observed from the ToF signals, as shown in the yellow patch in Fig. 3(a). To enable the model to recognize variation in ToF signals, the window size is set to 1.5 s, which is longer than the average duration of a gait cycle [49], [50]. The stride length between consecutive sliding windows in this work is determined from the system's sampling rate and the duration of the pre-impact falling period. A large window stride length causes the windows to skip over the pre-impact falling period which typically lasts from 200 to 400 ms in the fall trials [27]. Conversely, using a small window stride length raises the possibility of overfitting. To avoid overfitting and ensure that the window includes the pre-impact fall period for model training, the stride length is set to five sampling periods, which is equivalent to 167 ms.

E. Labeling Strategy

The traditional method for labeling the pre-impact fall period relies on video analysis or manual annotation. Due to the diversity of falling movements, defining the exact start timestamp of a pre-impact fall period is difficult, resulting in mislabeling. The proposed system utilizes ToF sensors and IMUs, allowing for precise timestamping of each type of activity. Fig. 3 presents the collected signals associated with four activities, along with corresponding labels. Each timestamp in the collected sequences is labeled with fall-risk levels and terrains for DTL. Three fall-risk levels, identified for each activity, are "normal," "near-fall," and "fall."

1) Normal: ADLs, such as walking, sitting, ascending stairs, descending stairs, and leg-shaking, are categorized as "normal."
2) Near-fall: Tripping and missteps are categorized as "near-fall." When a "near-fall" occurs, D_4, which represents the ToF signal from the heel of the shoe, yields nonzero ranging readings due to the heel-off movement. Hence, the start timestamp t_s of a near-fall movement can be determined. The end timestamp of a near-fall movement is the time when D_4 returns to zero, indicating that a heel strike has occurred, and the user's body is back in balance. After near-fall movement, the rest of the sequence is labeled "normal."
3) Fall: When a fall occurs, the ToF signal from the heel of the shoe is similar to that in a near-fall movement, where D_4 yields nonzero ranging readings and the start timestamp t_s of "pre-impact fall" is determined. The "pre-impact fall" ends with the fall impact, which is associated with the timestamp of the highest SMV value in the falling trial. After the fall impact, the rest of the sequence in each fall trial is labeled as "post-impact fall." This label is used to assess fall detection performance. The "fall" category includes both "pre-impact fall" and "post-impact fall."

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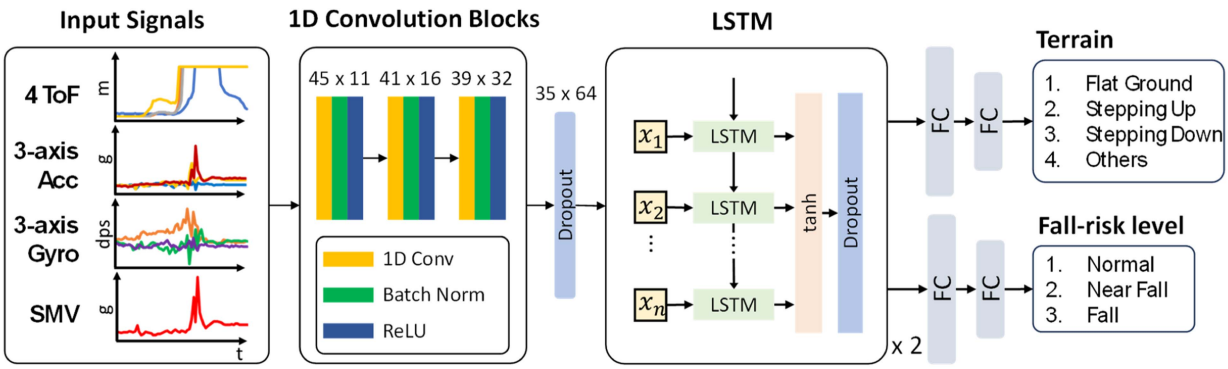


Fig. 4. Structure of DTL-CNNLSTM network for classifying terrain and risk of falling.

The early stages of "near-fall" and "pre-impact fall" exhibit similarities, promoting false alarms in fall-risk level classification. Thus, in this work, a constant $\tau \in [0, 1]$ is applied to modify the period length, as follows.

$$t'_s = t_s + \tau\,(t_e - t_s) \tag{2}$$

where t_e represents the end timestamps of "near-fall" and "pre-impact fall", and t'_s denotes the delayed start timestamp. Hence, the intervals between t'_s and t_e are used to label "near-fall" or "pre-impact fall" in the training of the model. This labeling strategy shortens the misleading periods associated with the DL model and increases the classification accuracy of the fall-risk levels.

Owing to the significant correlation between stepping ability and risk of falling [51], human stepping reactions are used in this work to categorize terrain as "flat ground," "stepping up," "stepping down," and "other." This labeling strategy enhances the interdependence between terrain and fall-risk classification tasks, enabling the model to achieve superior performance by using the proposed DTL architecture.

1) Flat ground: The label "flat ground" is associated with activities that are carried out on flat ground, such as walking, leg-shaking, and sitting. The trials do not require subjects to pass any obstacle, go upstairs, or go downstairs.
2) Stepping up: Crossing obstacles and ascending stairs are categorized under the "stepping up" terrain type. An obstacle or stair within 1 m of the front of the shoe can be identified by a ToF1 sensor. Based on the observed stance phase in ToF signals, the start of "stepping up" is time-stamped when D_1 is fixed and less than 1 m, as shown in Fig. 3(b) and (c). The end of "stepping up" is time-stamped at the fall impact or a return to flat ground.
3) Stepping down: Uneven floor and descending stairs are categorized under the "stepping down" terrain type. The start of "stepping down" is time-stamped at the heel-off movements before the descent on an uneven floor or stair, where D_4 exceeds zero. The end of "stepping down" is time-stamped at a return to flat ground or a fall impact.
4) Other: Foot movements after a fall impact differ from those observed during ADLs, causing the ToF sensors to fail to measure terrain geometry correctly. Hence, the post-impact fall period is labeled as "other" terrain to ensure that the proposed system accurately identifies the terrain.

Each timestamp is labeled with a fall-risk level and a type of terrain to train a DTL model. This labeling strategy enables the model to learn the relationship between terrain and fall-risk level. As a result, the model generates better feature representations in the latent space than traditional STL architectures for PIFD, favoring its general applicability across diverse terrains.

F. DTL-CNNLSTM Network

Fig. 4. presents the proposed DTL-CNNLSTM network structure. The segmented windows of four ToF signals, tri-accelerations, tri-angular velocities, and SMV are the inputs of this network. To improve system robustness, a hard parameter-sharing architecture of the DTL network is implemented. The concept of DTL originates from multi-task learning [52], [53], [54], [55], [56], [57]. DTL enhances the performance of related tasks by sharing learned features and utilizing information across tasks. The proposed DTL network consists of shared layers and task-specific layers. The shared layers form a modified CNNLSTM model, and the two task-specific layers that are based on fully connected (FC) layers are used to determine fall-risk level and terrain. The CNNLSTM model contains three one-dimensional convolution neural network (1D-CNN) blocks and two LSTM networks. Each 1D-CNN block consists of a 1D-CNN layer, a rectified linear unit (ReLU) activation function, and a batch normalization layer (Batch Norm) to extract temporal features from the time sequences across input channels. The function of a 1D-CNN layer is given by

$$x_j^l = b_j^l + \sum_{i=0}^{N^{l-1}} x_i^{l-1} \circledast w_{ij}^l \tag{3}$$

where x_i^{l-1} and w_{ij}^l are the input features and kernel of the ith neuron at layer l, respectively; $\circledast$ represents the convolution operation; N^{l-1} is the number of the input channels in layer l; b_j^l is associated bias, and x_j^l are the output features of the jth neuron in layer l. The 1D-CNN blocks output sequential features, which are then fed into the LSTM networks to capture long-term dependencies of these sequential features.

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The features that are extracted by the two LSTM networks are used for classification by task-specific layers, which are two FC layers with a softmax activation function for classification. Fall and near-fall sequences have only a specific segment that is labeled as "fall" or "near-fall," as shown in Fig. 3(b), (c), and (d). Hence, the number of windows labeled as "near-fall" or "fall" is smaller than those labeled as "normal," leading to the data distribution imbalance issue. To mitigate this issue, alpha-balanced focal loss, which is a dynamically weighted loss function that prioritizes difficult-to-classify windows, is implemented [58], [59]. Terrain-related loss L_{ter} and fall-risk level-related loss L_f are calculated from focal loss for DTL as, respectively,

$$L_{ter} = \begin{cases} -\alpha(1-p_{ter})^{\gamma}\log\left(p_{ter}\right), & y_{ter}=1 \\ -\left(1-\alpha\right)p_{ter}{}^{\gamma}\log\left(1-p_{ter}\right), & \text{otherwise} \end{cases} \tag{4}$$

$$L_f = \begin{cases} -\alpha(1-p_f)^{\gamma}\log\left(p_f\right), & y_f=1 \\ -\left(1-\alpha\right)p_f{}^{\gamma}\log\left(1-p_f\right), & \text{otherwise} \end{cases} \tag{5}$$

$$L_{total} = \lambda L_{ter} + (1-\lambda)\, L_f \tag{6}$$

where p_{ter} and p_f are the probabilities that are associated with terrain and fall-risk classes as predicted by the model, and y_{ter} and y_f are the class labels for both types of class. A weight factor $\alpha \in [0,1]$ is used to solve the class imbalance problem by assigning different weights to the loss function. This factor is based on the distribution of classes and penalizes the model more heavily when it misclassifies data from classes with less amount. γ is a focusing parameter, which is used to reduce the loss for well-classified examples. The total loss, denoted as L_{total}, is a weighted combination of the L_{ter} and the L_f. A weight factor $\lambda \in [0,1]$ determines the relative importance of each loss component. The proposed model is trained using the Adam optimizer with a learning rate of 10^{-3}. With this model training setup, total loss converges, and the model is trained to identify four terrains and three fall-risk levels from the multisensor signals.

G. Evaluation Metrics

This work utilized the leave-one-group-out cross-validation method for the proposed model. The 20 subjects were divided into ten groups, each consisting of one young subject and one elderly subject. The model was trained for each iteration using data from nine groups, while the remaining group served as the test set. A total of ten models were trained based on this validation approach, with each model making inferences on its corresponding test group. The overall performance was determined by aggregating the results across all test groups. This method validated the model's generalization to new subjects by systematically excluding one young subject and one elderly individual during each iteration, allowing the model to be tested on previously unseen data in people of different ages. All groups were tested, facilitating a comprehensive assessment of model performance for dual-task classification.

The window-based classification in terrain and fall-risk tasks is evaluated in terms of sensitivity (SEN), specificity (SPE), and accuracy (ACC). SEN represents the proportion of correctly identified windows that are contained in the target class, while SPE measures the proportion of actual non-target windows that are correctly identified. ACC represents the overall correctness of the model's predictions. The metrics SEN, SPE, and ACC are defined as

$$SEN = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{7}$$

$$SPE = \frac{\text{TN}}{\text{TN}+\text{FP}} \tag{8}$$

$$ACC = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{9}$$

where TP is the number of true positives, FP is the number of false positives, TN is the number of true negatives, and FN is the number of false negatives.

A sequence-based evaluation method is implemented to demonstrate the fall detection and prediction capabilities of the proposed model. Each sequence of test sets is segmented into consecutive windows, and the segments are fed into the trained model. If the model classifies three consecutive windows as associated with the same fall-risk level, then this level is regarded as the output of the model until another three consecutive windows have been thus classified to generate another output. The highest output fall-risk level within each sequence is the final sequence-based fall-risk classification result. A fall that is detected before the impact timestamp is a TP for both PIFD and fall detection (FD). If the fall is identified after the impact, it is considered to be an FN for PIFD but a TP for FD. This specification calculates the classification accuracy of different fall-risk level sequences. To evaluate the lead time (T_{lead}) of the PIFD, the T_{lead} of each successfully recognized PIFD fall sequence is given by

$$T_{lead} = t_{impact} - t_{pred} \tag{10}$$

where t_{impact} represents the timestamp of the fall impact, and t_{pred} denotes the first timestamp of a fall that is predicted by the model.

III. Results

The proposed system is validated using 1035 sequences that are obtained from ten young subjects and ten elderly individuals. These sequences are segmented into 139306 windows for model training and testing. The proportions of "normal," "near-fall," and "fall" fall-risk levels in the window-based dataset are 81.1%, 5.8%, and 13.1%, respectively. The terrains are 56.5% "flat ground," 17.1% "stepping up," 15.0% "stepping down," and 11.4% "other."

Table I presents the performance of classifying each segmented window using three fall-risk levels and four terrains. The results are based on inputs to the proposed DTL-CNNLSTM architecture, using IMU signals, ToF sensor signals, or a combination of both. ToF signals yield better terrain classification than single IMU signals, with 94.9% and 90.1% classification accuracies, respectively. In fall-risk level classification, using only IMU or ToF signals yields accuracies of 95.6% and 96.5%,

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TABLE I
COMPARISON OF SENSOR TYPES FOR WINDOW-BASED FALL-RISK LEVEL AND TERRAIN CLASSIFICATION RESULTS WITH HIGHLIGHTING FALL CLASSIFICATION PERFORMANCE IN PRE-IMPACT FALL PERIOD (UNIT = %)

	Terrain			Fall-Risk Level			**Pre-Impact Fall Period**		
	IMU	ToF	Both	IMU	ToF	Both	IMU	ToF	Both
SEN	87.0	92.4	**93.4**	84.4	88.9	**91.8**	79.5	63.0	**80.8**
SPE	95.4	97.7	**97.7**	94.4	95.2	**96.8**	98.8	98.8	**99.4**
ACC	90.1	94.9	**95.2**	95.6	96.5	**97.6**	94.0	89.9	**94.8**

TABLE II
SEQUENCE-BASED FALL-RISK LEVEL CLASSIFICATION RESULTS FOR DIFFERENT SENSOR TYPES (UNIT = %)

	Normal			Near Fall			PIFD			FD			Overall		
	IMU	ToF	Both	IMU	ToF	Both	IMU	ToF	Both	IMU	ToF	Both	IMU	ToF	Both
SEN	93.9	95.5	**96.5**	84.2	82.1	**97.5**	94.5	79.5	**95.2**	100.0	100.0	**100.0**	91.9	85.9	**96.2**
SPE	97.6	99.2	**99.5**	97.7	98.5	**98.5**	95.6	93.0	**99.3**	95.6	93.0	**99.3**	96.8	96.5	**99.2**
ACC	96.2	97.9	**98.5**	94.6	94.7	**98.3**	95.2	87.5	**97.7**	97.4	95.9	**99.6**	95.4	92.9	**98.1**

TABLE III
AVERAGE LEAD TIME (T_{LEAD}) AND LENGTH OF GROUND TRUTH PRE-IMPACT FALL PERIOD (T_{GT}) ON DIFFERENT TERRAINS

Metric \ Terrain	Flat Ground	Stepping Up	Stepping Down	Overall
T_{lead} (ms)	284.0	485.6	229.1	329.0
T_{GT} (ms)	397.1	630.6	302.2	440.3

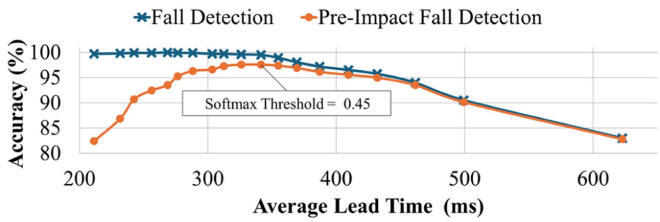


Fig. 5. Accuracies and average lead times for fall detection and PIFD across varying "fall" class softmax thresholds.

respectively. The highest accuracy in both tasks is obtained by using signals from both sensors as the inputs of the model, yielding 95.2% for terrain classification and 97.6% for fall-risk level classification. Comparing the results that are obtained using segmented windows provides evidence of the feasibility of the proposed system and data-processing flow.

To analyze further stability and robustness, the proposed model was applied to each collected sequence to detect the sequence-based results for each trial. Table II presents the overall sequence-based fall-risk level results. The proposed system detects near-fall movements with an accuracy of 98.3%. The accuracy of PIFD is 97.7%, indicating the ability of the system to predict accurately falls. All trial falls are detected by the proposed system, yielding a sensitivity of 100% and an FD accuracy of 99.6%. These results demonstrate that the proposed system accurately determines different fall-risk levels in each trial. This work collected 420 fall sequences, and the proposed model infers lead time for each fall sequence. Therefore, the average lead times from these sequences are used to validate the model. The proposed system detects each fall trial with an average lead time of 329ms, as shown in Table III. Notably, falls that occur while stepping down are associated with the smallest difference between predicted lead time and ground truth lead time, with means of 229.1ms and 302.2ms, respectively. The longest average ground truth lead time, 630.6ms, and average predicted lead time, 485.6ms, are obtained when stepping up. We attribute the difference in lead times between these two pre-impact fall movements to higher falling speed during downstairs movements, leading to a shorter duration between misstep and fall impact. Despite the limited time interval following misstep movements for classification, the proposed system accurately detects pre-impact falls.

This work classifies the signals into three fall-risk levels based on the highest softmax output value. To further optimize PIFD, we adjust the softmax threshold for the "fall" class probability. This threshold, varying from 0.05 to 0.95, determines the minimum probability that must be reached for the model to classify an input as associated with a fall class. Fig. 5. presents the impact of these thresholds on both fall classification accuracy and average lead time. At a lower threshold, the system exhibits greater sensitivity in detecting falls, leading to an increased rate of false alarms, with ADLs or near-fall movements classified as falls. As the threshold increases, fall detection accuracy improves. The threshold value optimally balances PIFD and FD accuracies, corresponding to approximately 330ms of average lead time. However, when the softmax threshold exceeds 0.45, the sensitivity of the system diminishes, impairing the system's ability to detect pre-impact fall movements reliably. These results show that using the highest softmax probability as the output of the three fall-risk levels and adjusting the softmax threshold both yield similar PIFD and FD performance.

To demonstrate the performance of the proposed system and its outperformance of the traditional STL architecture, seven other models are implemented, as shown in Fig. 6. These models are trained for the same number of epochs on the collected dataset. The traditional CNNLSTM (TCL) model, obtained from previous works [35], [36], [37], [38], [39], includes max-pooling layers in the CNN blocks to reduce the dimensionality of extracted features for the LSTM. However, in the TCL architecture, the extracted multi-sensor features share a limited latent space, affecting the data fusion performance. Removing the max-pooling layer can preserve these extracted features from

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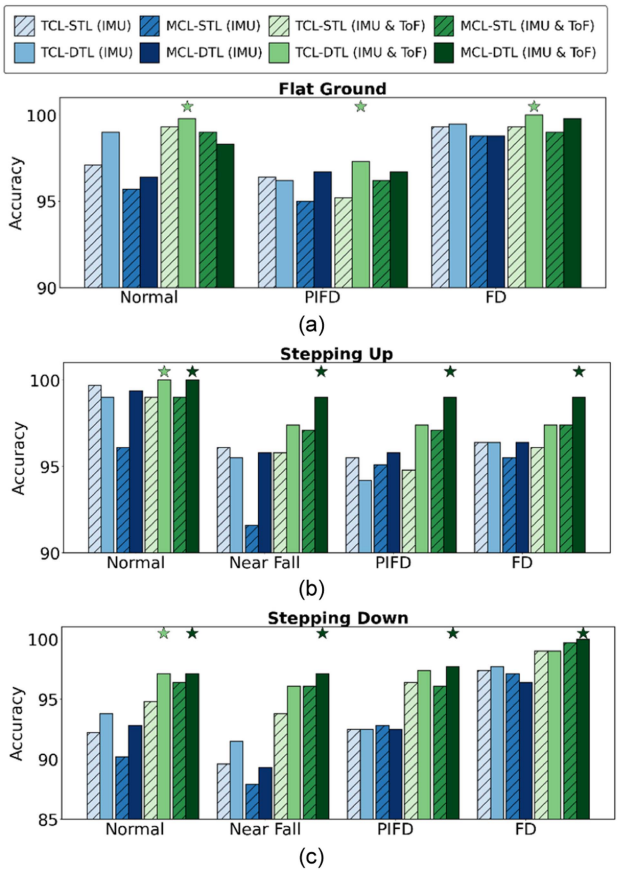


Fig. 6. Comparison of fall-risk level classification using different model architectures. Bars with a hatch pattern correspond to STL models, and bars without a hatch pattern correspond to DTL models. Blue bars indicate models based on IMU signals, and green bars indicate models based on both IMU and ToF signals. Stars indicate that model architecture is associated with highest fall-risk level classification accuracy among all models. Results based on activities on flat ground, stepping-up terrain, and stepping-down terrain with different fall-risk levels are shown in (a), (b), and (c), respectively.

multiple sensor signals during the fusion process in the CNN. Consequently, the proposed modified CNNLSTM (MCL) architecture eliminates the max-pooling layer in the CNN blocks. The results thus obtained demonstrate that TCL outperforms the MCL when only IMU data are used, with mean accuracies of 95.8% for TCL and 94.8% for MCL. However, when data from both sensors are used, MCL exhibits superior performance, achieving a mean accuracy of 98.0%, compared to 97.4% for TCL. In addition to comparing the designs of CNNLSTM models, this work reveals the impact of the DTL architecture on fall-risk level classification. The proposed method (MCL-DTL with ToF and IMU signals) improves overall fall-risk accuracy by 2.86% over that of the IMU-based TCL-STL method. Additionally, it achieves higher accuracies in near-fall detection and PIFD, which are 2.91% and 3.56% higher, respectively, on stepping-up terrain and 4.87% and 7.49% higher, respectively, on stepping-down terrains. These results confirm the improved fall-risk level classification performance of the proposed system across three terrains relative to those in previous works.

TABLE IV
LATENCY AND MODEL SIZE FOR DIFFERENT MODELS

Model	Latency (ms)		Model Size (KB)
	CPU	GPU	
TCL-STL	2.79	2.38	517.7
TCL-DTL	2.83	2.37	527.7
MCL-STL	4.36	2.48	615.2
MCL-DTL	4.50	2.66	903.8

Table IV presents the accuracy of the models with different architectures that are considered in this work and their corresponding latency measurements. The latency results represent the average computation time when the system runs each data in a test group on either the CPU or GPU, reflecting the efficiency of each model in detecting fall risk. The TCL models exhibit shorter response times than MCL models on the CPU and GPU. We attribute this result to the inclusion of max-pooling layers in the TCL models, which reduce the number of parameters and the overall model size. The GPU response times were measured at approximately 2.5ms per window with minimal variation across the models. The similar response times between the models are attributed to the fact that the computational capacity of the GPU substantially exceeds that required by the model. A latency comparison between the STL and DTL models reveals that scaling of the model minimally affects computation speed on the CPU and the GPU. The DTL approach does not introduce significant computational overhead, revealing that the system can handle multiple tasks simultaneously without sacrificing performance in real-time. In addition, the processing time of our model on a computer is less than the system sample rate, enabling the real-time assessment of fall-risk levels.

IV. DISCUSSION

The proposed system consists of a waist-worn device based on an IMU and a shoe-integrated device incorporating ToF sensors. ToF sensors generate ranging signals to detect terrain geometry, while IMU signals capture the user's motion. A modified CNNLSTM model is trained to process combined signals from both sensors as inputs. This model uses a DTL architecture, associating signals with three fall-risk levels - "normal," "near-fall," and "fall"- and four terrains - "flat ground," "stepping up," "stepping down," and "other." It determines the fall-risk levels associated with movements on various terrains, facilitating real-world fall-risk monitoring. To validate the proposed system, 16 activities are conducted by ten healthy young subjects in a laboratory setting. Additionally, ten elderly individuals perform four activities in a hospital setting. The results demonstrate that the proposed system and the DTL-CNNLSTM model can accurately recognize stepped terrains and different fall-risk levels while achieving PIFD.

Combining IMU and ToF sensors slightly improves the overall accuracy to 1.1% over that achieved using only the ToF sensor, as shown in Table I. Due to the imbalanced distribution of "near fall" and "fall" windows relative to "normal" windows, the contributions of the former two classes have a relatively minor impact on the final accuracy. This work uses a two-stage

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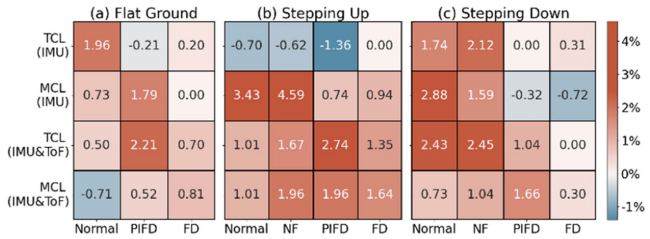


Fig. 7. Improvement of performance of DTL model over that of STL model at different fall-risk levels. Each value in grid represents difference between accuracies of STL and DTL models. Positive values (orange) indicate higher accuracy, while negative values (blue) indicate DTL model has lower accuracy.

evaluation method to incorporate window- and sequence-based results to obtain a comprehensive assessment. We posit that the ToF sensor cannot capture free-fall signals, unlike the IMU, thus affecting PIFD accuracy. Hence, the sensitivity of ToF signals in sequence-based PIFD is approximately 15.7% lower than that achieved with both sensors combined, as shown in Table II. Thus, combining IMU and ToF signals supports the proposed model in detecting pre-impact fall movements.

Fig. 6 reveals that in stepping-related regions, using both ToF sensors and IMU significantly improves performance over that achieved using a single IMU. The primary misclassification using the proposed model in stepping-up terrains is the identification of a "trip" as a "tripping fall." During balance recovery movements that are caused by a "trip," the IMU captures free-fall signals and high-SMV signals from the foot that has suddenly to support the body. These movements inhibit the accurate detection of pre-impact falls by the IMU-based system, generating false alarms. In trials on stepping-down terrains, the early stage of a misstep is similar to a movement that is associated with descending stairs, potentially causing the model to misclassify normal stair descent as a misstep, reducing the overall accuracy of the system. Hence, the results imply that relying on a single IMU to determine fall-risk levels on different terrains is less effective than using the proposed multisensory system, which can more accurately monitor fall risks.

Terrain classification and fall-risk level classification are regarded as two distinct tasks for a DTL architecture. DTL simultaneously improves the performance of both fall-risk level and terrain classification tasks by sharing learned representations between these tasks. After the model is trained, the model can infer fall-risk levels without the need for two FC layers to classify terrain, reducing the computational load. This model setting allows the system to operate with a response time equal to that achieved using STL during the practice application. Fig. 7 demonstrates DTL improvements in four pairs of models that have the same architecture and the same inputs but are associated with different sets of tasks. The difference between each pair is the input signals and the model's architecture. Eleven fall-risk trials on three terrains are conducted using the four pairs of models. The use of DTL improved the accuracy of all pairs of models in 33 trials, as shown in the orange grid-cells in Fig. 7. The results demonstrate that incorporating DTL into the models improved the detection of fall-risk movements on different terrains. DTL worsens the effectiveness of the IMU-based TCL model on stepping-up terrains. We attribute these results to the terrain classification capability of the model; the IMU-based TCL model less accurately recognizes stepping-up terrains than MCL models. The low accuracy of terrain classification results in a large total loss under the DTL setting, affecting its fall-risk level classification performance. However, the results suggest that incorporating the DTL architecture into models using MCL or both sensor types can effectively enhance performance across terrain trials. This improvement particularly demonstrates when the model has access to a greater variety of features, highlighting the benefits of using the proposed DTL architecture for the fall-risk level classification.

TABLE V
COMPARISON OF WINDOW-BASED FALL-RISK LEVEL ACCURACY ON DIFFERENT ENVIRONMENTS (UNIT = %)

	Flat Ground	Stepping Up	Stepping Down	Overall
Lab (Young Subject)	98.3	97.0	96.3	97.3
Hospital (Elderly)	99.9	99.9	97.2	99.8

To determine how environmental factors influence the movements of elderly people, the experimental protocol in a hospital involves different terrains with various elevation changes. Movements are associated with three fall-risk levels, which represent the balancing response of an elderly person to changes in terrain. However, hospital data collection involved real-world applications, such as those involving interference by another pedestrian and unfamiliar stairs. These conditions introduced variations in the data that differ from those in a controlled laboratory environment. As shown in Table V, the system achieved an overall accuracy of 99.8% in the hospital, and the accuracies of fall-risk level classification on flat ground, stepping-up terrain, and stepping-down terrain were 99.9%, 99.9%, and 97.2%, respectively. The results show that the proposed system was more accurate in the hospital than in the laboratory. We attribute this to the fact that young subjects conducted fall-related movements in the laboratory, and this involved extra complexity. The overall fall-risk level classification accuracies for all terrains across the two environments exceeded 97.3%, demonstrating the feasibility and reliability of the system in this work.

Table VI summarizes the performance of other PIFD methods, which are divided into the three categories of threshold-based methods, traditional machine learning approaches, and DL methods. Threshold-based methods have low power consumption but often yield a lower fall detection accuracy and offer shorter prediction lead times than other methods, limiting their effectiveness in complex situations [25]. Traditional machine learning methods can provide shorter response times, but they often suffer from overfitting, which limits their generalizability across different types of falls [26]. Most researchers have used DL models for PIFD [28], [30], [38], [39]. DL-based methods generally have better classification accuracy because DL models can capture nuanced patterns in sensor signals, enabling systems to classify similar but different movements effectively. However, works that have used DL-based methods have focused solely on fall-risk movements, neglecting the relevance of terrain to

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TABLE VI
COMPARISON OF PERFORMANCE WITH RELATED WORKS

Reference	2020 Wang [28]	2022 Ferreira de Sousa [25]	2022 Choi [30]	2023 Yu [38]	2023 Chai [26]	2024 Benoit [39]	**This work**
Sensor	IMU and Pressure Sensor	IMU	IMU	IMU	IMU	IMU	IMU and ToF Sensor
Placement of Sensor	Shoe Insole	Lower Back and Waist	Waist	Lower Back	Chest, 2 Wrists, 2 Elbows, 2 Thighs, and 2 Ankles	Lower Back	Waist and Shoe Outsole
Sample Rate	50 Hz	400 Hz	40 Hz	100 Hz	15 Hz	100 Hz	30 Hz
Number of Subjects	10	9	34	32	14	32	20 (10 elderly aged over 65)
Classification Types	4 Direction Falls and 5 ADLs	ADL and Fall	ADL, Near Fall, and Fall	ADL and Fall	ADL and Fall	ADL and Fall	**Fall-Risk Level**: ADL, Near Fall, and Fall **Terrain**: Flat Ground, Stepping Up/Down
Multi-terrain Analysis	No	No	No	No	No	No	**Yes**
Method	Multi-Source CNN Ensemble	Threshold-based Method	Modified DAG-CNN	CNNLSTM	Decision Tree	CNN DENSE	DTL-CNNLSTM
Performance	Acc.: 99.3%	Sens.: 94.0% Sepc.: 97.7%	Acc.: 99%	Balanced Acc.: 98.96%	Acc.: 97.9%	Acc.: 94.7%	Acc.: 98.1%
Lead Time (ms)	350	259	662	298.8	447.9	176.9	329

fall risk. This work addresses this issue by introducing DTL architecture to detect the fall-risk level and the terrain simultaneously. The results of this work show that using the DTL model improves PIFD performance by sharing learned representations between these tasks. In addition, this model architecture design increases fall-risk classification accuracy to 98.1%, particularly on varying terrain, confirming the feasibility and robustness of the system in real-world settings.

Some research explores the practical implementation of fall detection and prediction algorithms using edge systems [25], [26], [39]. However, PIFD methods for edge computing, such as threshold-based methods, traditional machine learning methods, or DL models, are necessary to balance power efficiency and performance based on the choice of edge computing ICs. This optimization is particularly crucial when DL models are applied in edge computing environments for fall detection, where device limitations demand a careful trade-off. Although running the developed model on an edge device is not the focus of this work, we fully recognize the importance of the development of edge computing for PIFD. Hence, we will consider insights related to edge computing in future works.

Elderly individuals have a slower fall reaction time than younger people, which inherently limits their ability to anticipate or avoid fall incidents [60]. Therefore, recent PIFD research has focused on ensuring the timely activation of wearable airbags, which typically require approximately 120 ms for total inflation [19], [20]. Table VI summarizes the lead times obtained in related works. Although the system in other listed PIFD works provides sufficient time for airbag inflation, the influence of terrain on lead time was not considered. This work is the first to analyze the lead time of PIFD on various terrains, as presented in Table III, thus proving the contributions of this work. Furthermore, this work introduces a terrain-robust DTL-CNNLSTM network for recognizing near-fall movements and falls on stepping terrains, which outperforms traditional IMU-based methods. The proposed system can be used to monitor elderly individuals who are unfamiliar with hospital environments or prone to falling on diverse terrains. Varying the terrain in training enables the system to be extended to other scenarios, such as long-term care facilities or home-care settings in the future. Such an extension can help to reduce falling-related injuries and thus enhance the safety of elderly individuals.

V. CONCLUSION

This work develops a wearable fall-risk monitoring system that classifies movements into three fall-risk levels - "normal," "near-fall," and "fall," taking into account the terrain. The wearable system consists of a waist-worn device and a shoe-integrated device, which synchronously generate an IMU signal and four ToF signals. All signals are input to a CNNLSTM model, which is designed with DTL architecture to perform terrain and fall-risk level classification tasks. The label "fall" in this work is associated with pre-impact and post-impact fall periods, enabling movements associated with a high risk of falling to be predicted and PIFD to be realized. Accuracies of 97.6% and 95.2% in classifying segmented windows with fall-risk levels and terrains, respectively, are achieved. Additionally, the sequence-based PIFD accuracy is 97.7%, with a fall prediction lead time of 329 ms, demonstrating the model's ability to accurately identify falls on various terrains. Ten elderly individuals are recruited to wear the proposed system and perform four ADLs. The overall monitoring accuracy for the elderly is 99.8%, revealing the general effectiveness of the proposed system. Considering environmental information, the proposed system has the potential to be extended to such contexts as long-term care facilities or home-care settings in the future.

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